

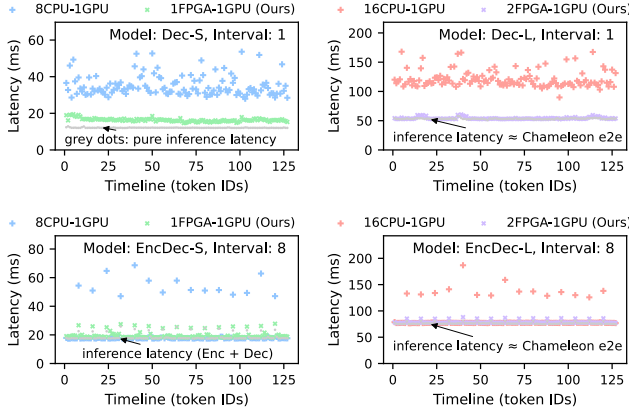


Figure 10: Latency of RALM inference given different LLM configurations and retrieval intervals.

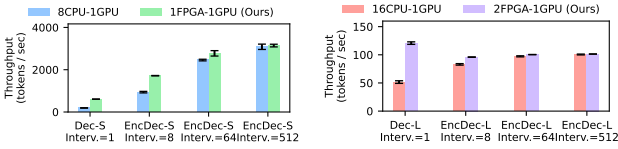


Figure 11: Throughput of RALM inference given different LLM configurations and retrieval intervals.

experiments), because, as the batch size grows, the latency increase for LLM inference is not as significant as that of vector search, due to the many-core parallelism that GPUs offer.

The need for resource disaggregation. Accelerator disaggregation allows Chameleon to adjust the ratio between the two types of accelerators across RALM configurations. We model the overall system throughput, measured by generated tokens per second, across various accelerator ratios using a total of 1,000 accelerators, assuming the cost for an inference accelerator and a retrieval accelerator is equivalent. Given retrieval interval i , batch size b , number of inference and retrieval accelerators N_I and N_R , latency per batch for inference and retrieval $L_I(b)$ and $L_R(b)$, the system throughput is determined by the minimum of the inference and retrieval throughput: $Th_{system} = \min(Th_I, Th_R)$, where $Th_I = \frac{i \cdot b \cdot N_I}{i \cdot L_I(b) + L_R(b)}$ and $Th_R = \frac{i \cdot b \cdot N_R}{L_R(b)}$. Figure 12 shows that the optimal ratio of accelerators to achieve the highest throughput varies significantly, ranging from 53.7%~99.0% across RALMs.

The disaggregated design, using the optimal accelerator ratio, consistently outperforms the monolithic ones with fixed ratios, as shown in Figure 13. Given the impracticality of adjusting the ratio for each RALM in a monolithic design, the performance of a monolithic design can only match that of Chameleon on a limited set of RALMs.

7 RELATED WORK

To our knowledge, Chameleon represents the first endeavor to improve RALM inference performance by using heterogeneous accelerator systems. We now introduce related research topics below.

ANN search. Researchers have developed various ANN search algorithms [14, 19, 30, 36, 54, 61, 75, 78, 84, 92, 93] and data management systems [22, 58, 62, 77, 79, 85]. Apart from PQ-based vector search, graph-based searching algorithms [17, 18, 53, 55, 56, 65, 81,

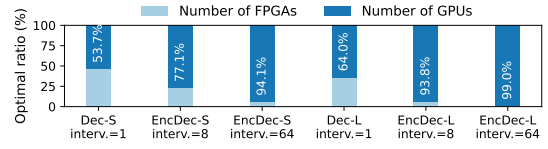


Figure 12: Disaggregation is essential as the optimal accelerator ratio varies significantly across RALM configurations.

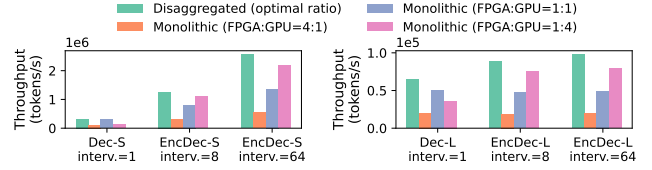


Figure 13: The disaggregated design consistently outperforms the monolithic ones using fixed accelerator ratios.

91, 94] are popular as they can achieve high recall and low latency. Locality-sensitive hashing (LSH) [15, 21] offers theoretical guarantees in ANN search but empirically does not perform as well as PQ and graph-based algorithms.

Vector search on modern hardware. Faiss is the most popular GPU-accelerated ANN search library so far [40], and there are several academic GPU implementations [10, 11, 52, 80]. Lee et al. [45] study ASIC designs for IVF-PQ, and a couple of works [38, 89] implement IVF-PQ on an FPGA, but their designs are constrained by either the limited HBM capacity or the slow CPU-FPGA interconnect. In contrast, Chameleon disaggregates IVF-PQ, with the index on GPUs and PQ codes on FPGA-based memory nodes, and employs the innovative hardware priority queue design to achieve high performance with little hardware resources. While graph-based vector search accelerators can achieve low latency [37, 88], the memory consumption is high, requiring up to one TB of memory for only one billion SIFT vectors, in contrast to 24 GB in our case. Apart from accelerators, researchers also study memory and storage for vector search. One can leverage non-volatile memory [71] and CXL [33] to scale up graph-based ANN, while on-disk ANN has to be more careful with I/O cost [9, 34, 47]. Hu et al. [28] further push down distance evaluation into NAND flash to reduce data movement.

8 CONCLUSION AND OUTLOOK

We present Chameleon, a heterogeneous and disaggregated accelerator system for efficient RALM inference. Given the rapidly evolving algorithms, software, and hardware related to RALMs, Chameleon can be potentially upgraded in the following ways. For LLM inference, ChamLM could be enhanced by supporting low precision [2], continuous batching [87], paged-attention [44], and disaggregated prompt computation and token generation [63]. While currently supporting PQ, ChamVS could potentially be replaced by graph-based ANN accelerators [64, 88]. ChamVS could also be extended to support index updates [83] and relational features [90].

ACKNOWLEDGMENTS

We thank AMD for their generous donation of the Heterogeneous Accelerated Compute Clusters (HACC) at ETH Zurich (<https://systems.ethz.ch/research/data-processing-on-modern-hardware/hacc.html>), on which the experiments were conducted.